

Module	Topic	Sub Topic
AIATB_01	AI Fundamentals	Introduction AI Automation Tester Blueprint Program Tools you'll need (ChatGPT, Claude, Gemini, n8n, Langflow, IDEs) What is Generative AI? How LLMs work (Tokenization, Context, Reasoning) GPT, Claude, Gemini, DeepSeek, vs Open Source Models LLM Comparison Sheet (PDF) Local Install LLM Private via Ollama (mistral, deepseek, GPT OSS) Difference Between ML, AI, Deep Learning
AIATB_02	Prompt Engineering	What is Prompt Engineering and why we need to learn it Steps to follow for Effective Prompt Engineering Role-based prompting for QA, Zero Shot vs Precise Prompting. Write your first QA-style prompt, Chain-of-Thought Prompt frameworks: STAR, CLEAR, CRISP Context creation via the Templates (.md files) RICE POT Framework (QA)
AIATB_03	Essential AI Tools Setup	ChatGPT Setup, Claude & Gemini Setup Claude desktop setup, LM Studio setup, Blackbox AI Using Ollama / LM Studio locally (optional) Anti Gravity GitHub Copilot Cursor for testers Checklist: AI Tools Setup Validation Augment, Github Copilt, Amazon Q, IntelliJ, Claude Code,
AIATB_04	Generation with AI for QA (Test Design, Strategy & Automation)	Requirement Analysis with AI Test Plan Generation (Tempaltes) , Test Statergy, AI-Generated Test Design Bug Report , Test Metrics Automation Code Generation with AI Cluade Code (AI Terminal for QA Task, Automation Frameworks) Augment Overview with Framework.
AIATB_05	AI Agents, n8n, LangChain, Langflow, MCP & RAG Automation	Introduction to AI Agents Tools, Actions, Memory, Reasoning Loops Safety & Guardrails

	n8n Automate QA Workflows		Install n8n (Local & VPS)
			Understanding AI Nodes
			Trigger, AI Agent, Vector embeddings, Memory store, Tools
			Build: Test Plan Generator Agent
			Build: Jira Ticket → Test Case Agent with RAG
			Assignment: Create your own custom AI QA workflow, MCP
	Langflow		
			What is Langflow
			Chains, Tools, Agents
			Build a QA RAG System
			Build a "Requirement Understanding Agent"
			AI-powered QA Assistant (Langflow), MCP
	MCP Server (Model Context Protocol)		
			Lesson: What is MCP?
			Lesson: Building local testing tools using MCP
			Lesson: Code Review Tool using MCP
			Lesson: API Testing Tool using MCP
			Activity: Connect MCP Tools to ChatGPT
			Mini-project: Automation Healing Agent via MCP Tools
			Build your own MCP in JS/TS
AIATB_06	Advanced AI Tools & Integrations		AI + Selenium (Selenium Full Project)
			AI + API testing workflows
			- Pojo Generation.
			- API Framework Setup "
			AI + Playwright (stagehand)
			AI + Appium (Vision detection)
	CI/CD + Reporting with AI		
			Jenkins pipeline + LLM integration (LLM, Jenkins), aws RUN
			GitHub Actions + AI summaries
			Failure clustering with AI (ReportPortal)

		Build "AI debugging step" in a pipeline
	LLM Testing (DeepEval), Trulens, PromptFoo, Open AI Eval	
		Why test LLMs?
		Accuracy, safety, reliability checks
		DeepEval setup + test writing
		Test hallucinations & structured outputs
		LLM Test Template
		Overview of DeepEval, PromptFoo, Trulens
		Live AI Projects & AI Testing Tool Creation
	Project 1: AI Test Plan Generator (RAG + Jira)	Build flow: Jira → RAG (Pinecone) → LLM → Test Plan
	Project 2: API Automation Agent	OpenAPI → Test Cases → REST Assured Scripts → Jenkins Execution
	Project 3: Visual Bug Analyzer	Screenshot → AI detection → Bug Report
	Project 4: AI-Enhanced Selenium Framework	Hybrid framework + AI locators + auto-healing
	Project 5: Your Own AI Full QA/Tester SaaS Tool	Req -> Test Plan -> Test Case Generator - Playwright TC -> Run -> Report